

ORIGINAL ARTICLE

A novel explainable AI framework for multi-disease ocular classification and diabetic retinopathy severity grading

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Abstract

This work introduces a two-stage explainable artificial intelligence framework for automated diagnosis of eye diseases. The objective is to support early screening and improve diagnostic accuracy by integrating multi-disease classification with severity grading of diabetic retinopathy. The proposed system is built on Vision Transformer models and enhanced through data augmentation, class balancing, and optimization techniques. In the first stage, the model classifies images into four categories: cataract, glaucoma, diabetic retinopathy, and normal. In the second stage, cases identified as diabetic retinopathy are further graded into early non-proliferative, severe non-proliferative, and proliferative stages. The performance of the presented architecture is evaluated using accuracy, precision, sensitivity, and F1-score. The results of the first stage shows that the Swin Transformer achieves an accuracy of 94.15%, with precision of 0.9415, sensitivity (recall) of 0.9404, and an F1-score of 0.9405. In the second stage, the Vision Transformer provides superior performance with an accuracy of 96.59%, precision of 0.9661, sensitivity of 0.9659, and an F1-score of 0.9658. In addition, the results of the ablation studies confirm that training strategies significantly influences the performance of the presented model. In order to increase transparency, clinical reliability, and interpretability, visualization techniques based on the attention mechanism are employed, which provide meaningful insight into the decision-making areas of the model. The results demonstrate that the proposed framework accurately classify multiple eye diseases and reliably assess the severity of diabetic retinopathy. The proposed approach has the potential to enhance early detection, reduce diagnostic errors, and increase trust in artificial intelligence-based tools in ophthalmology.

Keywords Convolutional neural networks · Deep learning · Diabetic retinopathy · Image classification · Medical imaging · Retinal images · Swin transformer · Transfer learning · Vision transformer

1 Introduction

Diabetes is a chronic metabolic disorder affecting a vast global population and is associated with serious complications, including heart disease, kidney failure, nerve damage, and visual impairment [1]. According to the International Diabetes Federation (IDF), the number of people living with diabetes is expected to reach 643 million by 2030 [2]. Among the most vision-threatening complications is diabetic retinopathy (DR), a progressive



eye disease that can lead to irreversible vision loss if undetected and untreated [3]. The global rise in diabetes, particularly in low- and middle-income countries, has significantly increased the prevalence of DR, making it a major contributor to the global burden of blindness [4, 5].

While DR is the most common ocular complication of diabetes, other eye diseases such as cataract and glaucoma also contribute substantially to visual impairment worldwide. Cataracts involve lens opacification and are primarily age-related, though they may also arise from trauma, metabolic disorders, or hereditary factors. Cataract surgery remains one of the most successful interventions to restore sight [6]. In contrast, glaucoma is a group of optic neuropathies often linked to elevated intraocular pressure (IOP), and it progresses insidiously until significant and irreversible vision damage occurs [7]. Early detection via regular eye screening is essential to prevent vision loss from glaucoma.

Among diabetic individuals, DR poses a significant threat to vision. In 2021, an estimated 9.6 million people in the United States alone, accounting for 26.4% of those with diabetes, had DR, with 1.84 million experiencing vision-threatening diabetic retinopathy (VTDR) [8]. DR occurs when prolonged high blood sugar damages retinal blood vessels, leading to leakage, hemorrhage, and abnormal neovascularization [9]. The condition progresses through three primary stages: early non-proliferative diabetic retinopathy (NPDR) marked by microaneurysms and mild edema; severe NPDR, characterized by extensive ischemia and vascular blockage; and proliferative diabetic retinopathy (PDR), where fragile new vessels may cause vitreous hemorrhage, fibrosis, and potential retinal detachment [10–12]. Early-stage DR is often asymptomatic, making routine screening crucial for timely intervention and vision preservation. Effective and scalable screening methods are vital, especially in underserved regions where healthcare infrastructure and ophthalmologist availability are limited [13, 14]. Conventional diagnostic techniques, such as fundus photography, optical coherence tomography (OCT), and fluorescein angiography, still require trained ophthalmologists to interpret results, creating bottlenecks in DR diagnosis and treatment [15, 16].

To address this, automated DR detection using artificial intelligence (AI) has gained attention. However, current approaches, primarily based on convolutional neural networks (CNNs), face challenges in analyzing retinal images due to their inability to model long-range dependencies and global contextual features [17, 18]. These limitations are especially critical when detecting subtle or complex lesions in advanced DR stages. Moreover, CNN-based models often lack interpretability, making them “black-box” systems and hindering their acceptance in clinical settings where transparency and explainability are essential for trust and adoption [19, 20].

Recently, Vision Transformers (ViTs) have emerged as a compelling alternative for medical image analysis. Unlike CNNs, ViTs divide images into non-overlapping patches and process them as sequences using self-attention mechanisms—enabling the extraction of both local and global features critical for accurate classification [21–23]. ViTs have demonstrated superior performance in detecting high-resolution and complex patterns, making them well-suited for retinal image analysis, including DR detection [24, 25].

Another driving force behind this research is the increasing demand for Explainable AI (XAI) in healthcare. XAI techniques aim to improve model transparency by offering visual and feature-based rationales for decisions, fostering clinician confidence and facilitating integration into real-world workflows [26–28].

Eye disease detection using deep learning techniques has gained significant attention, particularly for classifying conditions like normal, cataract, glaucoma, and diabetic retinopathy from retinal images. Several studies have explored various deep learning architectures to improve the accuracy and efficiency of automated classification. One such study applied the EfficientNet-B0 architecture to classify eye diseases using a dataset of 4217 retinal images, divided into training, testing, and validation sets. The preprocessing steps involved resizing, cropping, normalization, and one-hot encoding. With the inclusion of data augmentation and hyperparameter optimization, the model achieved an accuracy of 86%, with precision, recall, and F1-score all reaching 87% [29]. This result suggests that the EfficientNet-B0 model offers a promising solution for classifying common eye diseases.

In a similar context, another study utilized a custom convolutional neural network to classify retinal images into the four categories of normal, cataract, glaucoma, and diabetic retinopathy. The dataset, consisting of 4217 images, was optimized using the Adam optimizer, with a training protocol set for 15 epochs and a batch size of 32. The CNN model yielded a high accuracy of 94%, indicating that the model was highly effective in distinguishing

between these eye conditions [30]. This approach emphasizes the importance of deep learning for accurate and timely detection of eye diseases, providing a reliable tool for clinical use.

Transfer learning combined with CNNs was explored in another study to address the challenges of diagnosing eye diseases, using a dataset of 4217 images. The model achieved an impressive accuracy of 95%, with precision and recall rates notably high, especially for Diabetic Retinopathy (100%). The study suggested that expanding the dataset to include a broader range of ocular diseases could further improve the model's performance and help clinicians with early intervention [31]. This demonstrates the potential of deep learning techniques to enhance diagnostic accuracy in ophthalmology.

The EfficientNetB3 model was also applied in a study to classify normal, cataract, glaucoma, and diabetic retinopathy conditions using retinal images. The model was trained with data augmentation, class weighting, and fine-tuning, achieving a classification accuracy of 93%. Notably, the model performed exceptionally well in identifying diabetic retinopathy and cataract, further highlighting the capacity of deep learning models to assist ophthalmologists in early detection and improved diagnostic accuracy [32]. The results indicate that EfficientNetB3 strikes a strong balance between accuracy and computational efficiency, making it a suitable choice for clinical applications.

In another comparative study, several deep learning models, including DenseNet, EfficientNet, Xception, VGG, and ResNet, were evaluated on a dataset of 4217 retinal images for classifying the same four conditions. Among these models, EfficientNet demonstrated superior performance, delivering excellent results across various evaluation metrics, such as accuracy, precision, recall, and F1-score. The study emphasized that EfficientNet outperformed other models like VGG and ResNet, making it an ideal candidate for retinal disease classification [33]. These findings underline the effectiveness of EfficientNet in handling retinal image classification tasks.

Finally, another study [34] proposed a weighted ensemble deep learning approach combined with Two-Dimensional Discrete Wavelet Transform (2D DWT) and 2D principal component analysis (PCA) for feature extraction and dimensionality reduction. Pre-trained models like EfficientNetB6 and DenseNet169 were evaluated for eye disease detection. The ensemble method achieved 94.9% accuracy with 67.3% data reduction using a single level of 2D DWT, and 92.9% accuracy with 93.75% reduction when both two-level 2D DWT and PCA were applied. These results outperformed prior work on the same Kaggle dataset, demonstrating the effectiveness of ensemble learning and dimensionality reduction for enhancing DL performance.

While the aforementioned studies focus on classifying ocular diseases, deep learning has also shown promise in predicting systemic conditions such as diabetes [35]. This highlights the expanding scope of AI applications in broader medical diagnostics and risk stratification.

A wide range of techniques has been employed for diabetic retinopathy classification, including machine learning, image processing, and deep learning methods such as convolutional neural networks. These approaches have utilized handcrafted features, ensemble models, lesion segmentation, and data augmentation to improve accuracy and robustness. Despite notable success, limitations in capturing global image context and handling class imbalance remain challenges.

Recent studies have increasingly explored vision transformer-based models for the classification of diabetic retinopathy, with promising results. A study by [36] proposed a vision transformer-based method for DR severity classification, leveraging non-overlapping patches of fundus images to retain spatial information, which are then processed by multi-head attention layers. The model demonstrated superior performance over traditional convolutional neural networks and other state-of-the-art methods, achieving an accuracy of 87.41%. The classification results were notable, with Grade 0 achieving 97.33% accuracy and Grade 1 reaching 93%. However, the study faced limitations due to the relatively small dataset of 2,500 images, which raises concerns about the model's generalizability. Additionally, the absence of key evaluation metrics, such as the confusion matrix, limits the interpretability of misclassification patterns.

In another study [37], introduced a ViT-based deep learning pipeline for DR severity classification, utilizing the newly released FGADR dataset to address the need for large-scale data in transformer models. The study incorporated data balancing techniques like focal loss, label smoothing, and class weights, achieving an F1-score

of 0.825 and an AUC of 0.964. Despite its advancements, the model exhibited relatively low accuracy, indicating a need for further optimization. Future improvements could focus on enhancing robustness through external validation, lesion segmentation, and bias assessment.

Expanding the application of ViTs [38], explored the use of wide-field optical coherence tomography angiography (WF-OCTA) images for automatic DR diagnosis. The model, which classified DR into four severity stages, achieved an impressive accuracy of 99.20%. However, the small dataset of 288 diabetic patients and 97 healthy individuals limited the generalizability of the results, requiring further validation on larger, more diverse datasets.

A hybrid deep learning model, ResViT FusionNet, proposed by [39], aimed to enhance DR classification by combining the strengths of ResNet50 and ViT. The model achieved a precision of 0.9307, a recall of 0.9300, and an F1-score of 0.9275, with data augmentation techniques improving robustness. Notably, explainable AI methods like LIME and Grad-CAM were employed to enhance transparency. Despite these successes, the study revealed a low recall of 0.664 for severe DR cases, suggesting that the model may miss critical diagnoses. Moreover, the lack of computational details and reliance on a single dataset limit the model's generalizability and practical deployment.

In [40], a hybrid model combining a fine-tuned vision transformer and a modified capsule network (CapsNet) was proposed for DR severity prediction. The model incorporated preprocessing techniques such as power law transformation and contrast-limiting adaptive histogram equalization, achieving high accuracy across four datasets, including 88.18% on AP-TOS and 87.78% on Messidor-2. However, the model underperformed on the EyePACS dataset, with poor precision, recall, and F1-score, suggesting limited generalization. Furthermore, the increased training time due to CapsNet's routing algorithm and the lack of model explainability are notable drawbacks.

A ViT-based method incorporating an enhanced multi-head self-attention mechanism and a Linformer backbone for improved computational efficiency was presented in [41]. The model processed non-overlapping image patches and achieved an accuracy of 88.13% on the IDRiD dataset. However, the small dataset size raised concerns about its generalizability to larger, real-world datasets.

In [42], a multi-headed CNN and ViT-based model was proposed for DR severity classification, where multiple CNN heads extracted local features, which were processed by ViT for both local and global spatial information. The model achieved an impressive classification accuracy of 96.0%. However, the lack of details regarding the dataset sources, augmentation techniques, and training parameters limits the generalizability and applicability of the model.

The diabetic retinopathy compact convolutional transformer (DRCCT) model proposed by [43] combined convolutional layers with transformer techniques for improved classification across five DR categories. The model achieved 96.93% testing accuracy, with an F1-score of 96.96% and a recall of 98.89%, demonstrating an effective balance between false positives and false negatives. However, challenges such as imbalanced datasets, particularly distinguishing between mild and moderate stages, were noted. Oversampling was used to address class imbalance, though it may have led to overfitting, and inconsistencies in dataset reporting raised concerns about data handling.

A Swin Transformer-based framework for automatic DR stage detection was introduced in [44], utilizing Contrast Limited Adaptive Histogram Equalization (CLAHE) and data augmentation techniques. The model achieved 92.1% accuracy on the APTOS 2019 Blindness Detection dataset, with a sensitivity of 94.01% and an F1-score of 93.89%. However, the lack of a validation set during training raises concerns about the model's generalizability, and the model was evaluated only on a single dataset, limiting its broader applicability.

Finally, authors in [45] proposed a ViT based method for DR grade recognition, demonstrating that attention mechanisms can replace CNNs for this task. The model achieved an accuracy of 91.4%, with a specificity of 0.977 and a precision of 0.928. However, the study was limited by the use of a single public dataset and the high computational cost of the ViT-L model, which requires 37 h of training and 307 million parameters. Moreover, class imbalance remained a significant issue, which could result in biased performance.

Most existing work focus solely on DR detection, often ignoring other common retinal diseases such as cataract and glaucoma. To date, no models integrate both multi-disease classification and DR severity grading in a unified framework; instead, most existing approaches focus on a single task and rely heavily on CNNs with limited interpretability. Moreover, despite significant advances in the application of deep learning methods to retinal fundus image analysis, a significant portion of existing systems are still far from real clinical needs. Many previous studies have focused on the diagnosis of a specific disease and are therefore unable to support practical screening scenarios, where the simultaneous differentiation between multiple eye diseases such as cataract, glaucoma, and DR is essential. This limitation reduces the clinical applicability of existing models, especially in resource-limited screening settings. Furthermore, even in studies that have addressed diabetic retinopathy, the focus has often been either solely on the diagnosis of the disease or its severity grading has been performed separately and in independent frameworks. This separation of tasks not only leads to increased system complexity and computational costs, but also prevents the creation of an integrated decision-making flow that is consistent with clinical processes. From a technical perspective, although deep models have achieved high accuracies, the lack of clinical interpretability remains a major obstacle to their adoption in medical settings. The lack of a clear explanation of the decision-making areas of the model reduces the confidence of clinicians and challenges the practical use of these systems. This is especially important in sensitive applications such as screening and monitoring the progression of diabetic retinopathy.

In response to these clinical and technical gaps, the present study presents a two-stage explainable AI-based framework that integrates the simultaneous diagnosis of multiple eye diseases with the grading of diabetic retinopathy severity in a single integrated system. By purposefully utilizing Transformer-based architectures and attention-based techniques, this framework not only achieves accurate and stable performance, but also increases the transparency of model decisions, taking an effective step towards the convergence of technical developments with clinical practical requirements. This paper proposes a novel end-to-end deep learning framework based on ViTs that simultaneously performs multi-disease classification (DR, cataract, glaucoma, or normal) and, if DR is detected, automatically stages its severity (early NPDR, severe NPDR, or PDR). The model is designed to be scalable, interpretable, and clinically applicable, particularly in low-resource clinical settings.

The key contributions of this paper can be summarized as follows:

- A unified, two-stage AI-based diagnostic framework for automated classification of major ocular diseases (cataract, glaucoma, and diabetic retinopathy) and fine-grained DR severity grading, tailored for real-world screening applications.
- The first application of vision transformers for multi-class ocular disease detection demonstrating superior performance over CNN-based and hybrid state-of-the-art methods across accuracy, F1-score, and recall.
- Integration of explainable AI techniques, including attention maps and saliency-based visualizations, enabling interpretable predictions that support clinician trust and diagnostic validation.
- Thorough ablation studies and robustness analyses validating the impact of preprocessing, augmentation, and class balancing strategies, with measurable improvements in accuracy and F1-score.
- A deployment-ready design focused on real-world usability, with a lightweight inference process, compatibility with non-GPU hardware, and adaptability to resource-limited clinical environments such as mobile screening units and rural healthcare settings.

2 Methods

This section outlines the methodology employed to develop and evaluate the proposed automated retinal disease diagnosis system. It includes details on the dataset, preprocessing techniques, model architecture, training strategies, and evaluation metrics. All experiments were conducted on a local Windows 10 machine equipped with an Intel64 Family 6 Model 94 Stepping 3 processor (4 physical cores, 8 logical threads) and 32 GB of RAM. The

training was performed using Python 3.11.9, PyTorch 2.4.1 (CPU-only), and Transformers 4.46.2 without GPU acceleration.

In our previous research [46], we investigated the use of ViT for binary classification tasks in retinal imaging. Specifically, we employed the vit-base-patch16-224-in21k model, pre-trained on ImageNet-21k, to classify fundus images into two categories: diabetic retinopathy versus normal. The results demonstrated that ViT models, with their self-attention mechanisms capable of capturing global contextual information, outperformed traditional CCNs in terms of robustness and generalization across variations in retinal imaging.

2.1 Dataset description and preprocessing

The first stage of the system involved training on a curated dataset comprising 4,217 retinal fundus images, evenly distributed across four diagnostic categories: Normal, Diabetic Retinopathy, Cataract, and Glaucoma. These images were collected from multiple publicly available sources, including the Indian Diabetic Retinopathy Image Dataset (IDRiD), the High-Resolution Fundus (HRF) dataset, and the Ocular Disease Recognition dataset, accessed via Kaggle [47].

To ensure consistency and robustness, all images were preprocessed by resizing them to 224×224 pixels and normalizing pixel values using the mean and standard deviation values of the Swin Transformer's pretrained processor. During training, images were augmented using a combination of probabilistic Contrast Limited Adaptive Histogram Equalization (CLAHE), geometric transformations (rotations, flips, affine transformations), and photometric adjustments (color jittering and random erasing). These augmentations aimed to enhance contrast, reduce illumination artifacts, and simulate real-world variability, improving the model's generalization capabilities. The key details of these preprocessing steps are outlined in Table 1, which summarizes the specific augmentation parameters used for stage 1.

The second stage focused on classifying the severity of diabetic retinopathy using the DDR (Diabetic Retinopathy Detection) dataset, which contains over 12,000 retinal images collected from hospitals across China. To streamline the classification task and focus exclusively on pathological cases, images labeled as "normal" were removed, yielding a subset of 6,256 images representing progressive stages of DR. These were re-labeled into three clinically meaningful categories: Early Non-Proliferative, Severe NPDR, and Proliferative DR, with early-stage labels consolidated to reduce inter-category ambiguity and simplify the model's decision boundaries.

To address the significant class imbalance, the dataset was balanced to include approximately 1,500 images per class. Overrepresented classes were reduced via random undersampling, while underrepresented classes were expanded using controlled oversampling combined with comprehensive data augmentation. The augmentation pipeline included transformations such as RandomResizedCrop, RandomAffine (incorporating rotation, translation, scaling, and shear), RandomHorizontalFlip, and GaussianBlur. This combined approach introduced sufficient variability to reduce the risk of overfitting to duplicated samples while preserving class-discriminative features. As a result, the model was exposed to a more diverse and balanced training set, improving both generalization and stability during learning. Table 2 lists the augmentation techniques applied in stage 2, which included transformations such as RandomAffine, RandomHorizontalFlip, ColorJitter, and RandomErasing. The training set was split into 70% for training, 15% for validation, and 15% for testing, using stratified sampling to preserve the class distribution across all subsets. The balanced dataset was split into training (70%), validation (15%), and test (15%) sets using stratified sampling to preserve class proportions across all subsets.

Table 1 Data augmentation techniques for stage 1

Augmentation Type	Feature Value
RandomResizedCrop	scale=(0.8, 1.0)
RandomAffine	degrees=15, translate=(0.1, 0.1), scale=(0.9, 1.1), shear=10
RandomHorizontalFlip	$p=0.5$
GaussianBlur	kernel_size=3, sigma=(0.1, 2.0)

Table 2 Data augmentation techniques for stage 2

Augmentation Type	Feature Value
RandomAffine	degrees=15, translate=(0.1, 0.1), scale=(0.8, 1.2)
RandomHorizontalFlip	RandomHorizontalFlip
RandomVerticalFlip	RandomVerticalFlip
RandomRotation	RandomRotation(30)
ColorJitter	brightness=0.3, contrast=0.3, saturation=0.3, hue=0.02
RandomErasing	$p=0.3$, scale=(0.02, 0.33), ratio=(0.3, 3.3)

Table 3 Model architecture and parameters

Parameter	Stage 1: Ocular Disease Detection	Stage 2: DR Severity Grading
Model Architecture	Swin transformer	Vision transformer
Pretrained Backbone	Swin-base-patch4-window7-224	ViT-base-patch16-224-in21k
Input Image Size	224 × 224	224 × 224
Number of Classes	4 (Cataract, DR, Glaucoma, Normal)	3 (Early NPDR, Severe NPDR, Proliferative DR)
Batch Size	16	32
Epochs	40	40
Optimizer	AdamW	AdamW
Learning Rate	1e-4	3e-5
Weight Decay	1e-2	1e-4

2.2 Model architectures and training

The primary classification architecture employed in the first stage was based on the Swin Transformer, specifically the *swin-base-patch4-window7-224* model pretrained on ImageNet-21k. The Swin backbone was followed by a dropout layer and a fully connected classifier mapping to four output classes. The entire Swin Transformer was fine-tuned end-to-end, allowing the model to adapt its feature representation to the medical imaging domain. Input images were normalized using the same mean and standard deviation values used during the model's original pretraining. The network was trained using the AdamW optimizer with a learning rate of 3e-5, weight decay of 1e-4, and a cosine annealing scheduler. Cross-entropy loss with label smoothing (set to 0.1) was employed to improve generalization and prevent overconfidence in predictions.

In the second stage, a Vision Transformer (*vit-base-patch16-224-in21k*) served as the primary model for classifying the relabeled diabetic retinopathy stages. This model, pretrained on ImageNet-21k, was fine-tuned to output three classes corresponding to the new DR stage labels. A linear classification head was appended to the ViT's CLS token output for final prediction. All layers of the model were fine-tuned using the AdamW optimizer with an initial learning rate of 1e-4 and weight decay of 0.01. A ReduceLROnPlateau scheduler, monitoring validation loss, was applied to adapt the learning rate and improve convergence stability. These architectural and training details are summarized in Table 3.

The use of different architectures in different stages of the proposed framework was a conscious decision based on the distinct nature of the tasks at each stage. In the first stage, the main goal is to simultaneously detect multiple eye diseases including cataract, glaucoma, and diabetic retinopathy from retinal fundus images; problems that are mainly associated with relatively distinct structural patterns and obvious differences in the spatial distribution of features. In this stage, the Swin Transformer model was chosen due to its hierarchical structure, local attention windows, and effective ability to extract multiscale features, as these features allow for simultaneous learning of local patterns and global relationships with high computational efficiency.

In contrast, the second stage focuses on the grading of diabetic retinopathy severity, which is an inherently delicate, continuous, and progressive problem that requires the detection of very subtle changes in vessels, retinal lesions, and scattered pathological areas. In this context, Vision Transformer shows a greater ability to focus on sparse and non-local patterns due to its global attention mechanism and its high capacity to model long-range

dependencies. The obtained experimental results, together with component elimination studies, confirm that this architectural separation leads to significant improvements in the accuracy, stability, and clinical interpretability of the system in both stages, strengthening the overall performance of the proposed framework.

2.3 Baseline CNN models

To evaluate the relative performance of the ViT architecture, the proposed model was benchmarked against traditional convolutional neural network architectures commonly used in medical imaging. The selected baselines included ResNet-50, VGG-16, EfficientNet-B0, and AlexNet. All models were trained under identical experimental conditions, using the same pre-processing steps, balanced datasets, and evaluation metrics.

2.4 Evaluation metrics

To assess the classification performance of the proposed models, a set of standard evaluation metrics was employed, including accuracy, precision, recall, F1-score, and ROC-AUC. These metrics were chosen to ensure a balanced evaluation, particularly in the context of class imbalance inherent to diabetic retinopathy datasets.

Accuracy is defined as the ratio of correctly predicted samples to the total number of samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

Precision quantifies the proportion of positive identifications that were actually correct and is computed as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Recall, also known as sensitivity, measures the proportion of actual positives that were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

The *F1-score* is the harmonic mean of precision and recall, providing a single measure that balances both concerns:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

To address class imbalance, macro-averaging was applied for precision, recall, and F1-score, treating each class equally regardless of sample count.

Finally, the Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) was employed to evaluate the model's overall discriminative capability in distinguishing between the different DR stages. For multi-class classification, ROC-AUC was computed using a one-vs-rest strategy, where a separate ROC curve is generated for each class against all others, and the average AUC is reported.

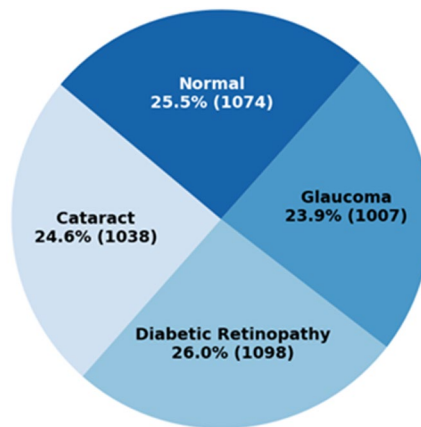


Fig. 1 The class distribution of the ocular diseases dataset

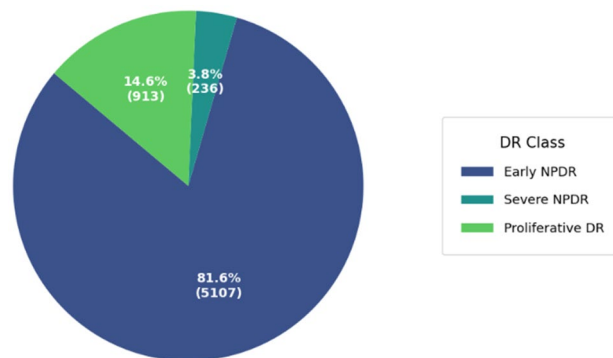


Fig. 2 The original class distribution of the DR severity dataset

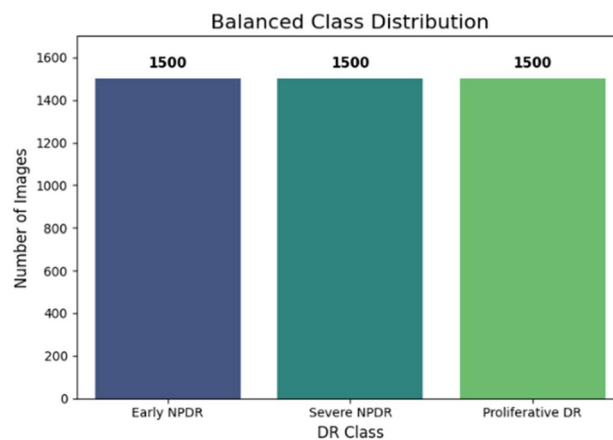


Fig. 3 The balanced class distribution of the DR severity dataset

Table 4 Classification report of the ocular disease classifier (stage 1)

Class	Precision	Recall	F1-Score	Support
Cataract	0.9434	0.9615	0.9524	156
DR	0.9702	0.9879	0.9790	165
Glaucoma	0.9429	0.8742	0.9072	151
Normal	0.9096	0.9379	0.9235	161
Accuracy	0.9415			633
Macro Avg	0.9415	0.9404	0.9405	633
Weighted Avg	0.9417	0.9415	0.9412	633

Table 5 Classification report of the DR severity classifier (stage 2)

Class	Precision	Recall	F1-Score	Support
Early NPDR	0.9724	0.9378	0.9548	225
Severe NPDR	0.9615	1.0000	0.9804	225
PDR	0.9643	0.9600	0.9621	225
Accuracy	0.9659			675
Macro Avg	0.9661	0.9659	0.9658	675
Weighted Avg	0.9661	0.9659	0.9658	675

3 Results

This section presents the experimental findings from both stages of the proposed diagnostic framework. The results include classification performance metrics, confusion matrices, ROC-AUC curves, and training and validation trends.

3.1 Dataset distribution and preprocessing summary

This subsection presents the class distribution of the two datasets utilized in the two-stage classification pipeline. In the first stage, the dataset comprised 4,217 retinal fundus images distributed relatively evenly across four diagnostic categories: Normal (25.5%), Diabetic Retinopathy (26.0%), Cataract (24.6%), and Glaucoma (23.9%), as shown in Fig. 1. This near-balanced distribution helped minimize class bias during training, enabling the initial classifier to learn generalized feature representations across diverse ocular pathologies.

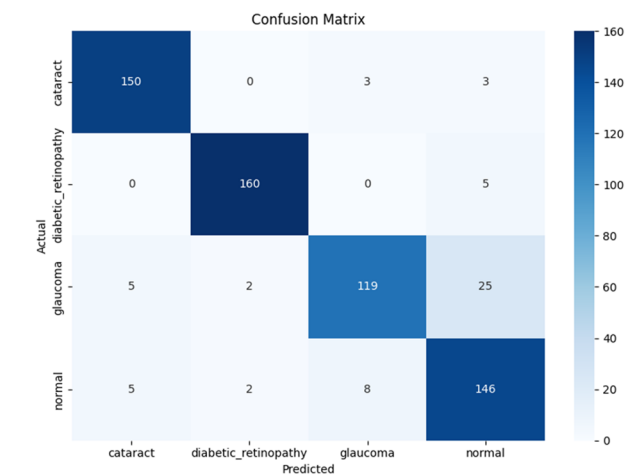
In contrast, the second stage focused exclusively on DR severity classification, which initially presented a marked class imbalance. As shown in Fig. 2, the original DR subset consisted predominantly of Early Non-Proliferative DR (81.6%), with far fewer instances of Severe NPDR (14.6%) and Proliferative DR (3.8%). Such skewed distribution poses a risk of model bias and reduced sensitivity to clinically critical but underrepresented stages.

To mitigate this, the dataset was rebalanced through a combination of random under-sampling of the majority class and synthetic oversampling of minority classes, resulting in a uniform distribution of approximately 1,500 images per class as illustrated in Fig. 3. This balancing was further reinforced by an extensive data augmentation pipeline tailored to preserve pathological characteristics while enhancing sample diversity. These measures were crucial in ensuring robust model training and equitable performance across all DR severity levels.

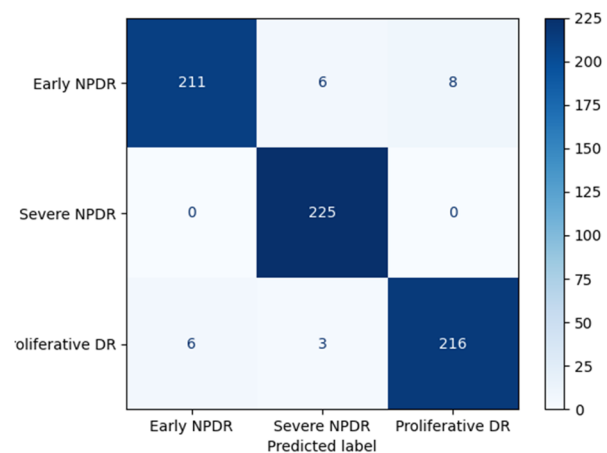
3.2 Model performance on test set

The performance of the proposed two-stage classification framework was quantitatively assessed using stratified test sets, each comprising 15% of the respective datasets. Classification results were evaluated using standard metrics: Precision, Recall, F1-Score, and Support, providing insights into both class-specific and overall predictive performance.

Fig. 4 (a) Confusion matrix of ocular disease classifier. **(b)** Confusion matrix of DR severity classifier



(a). Confusion matrix of ocular disease classifier.



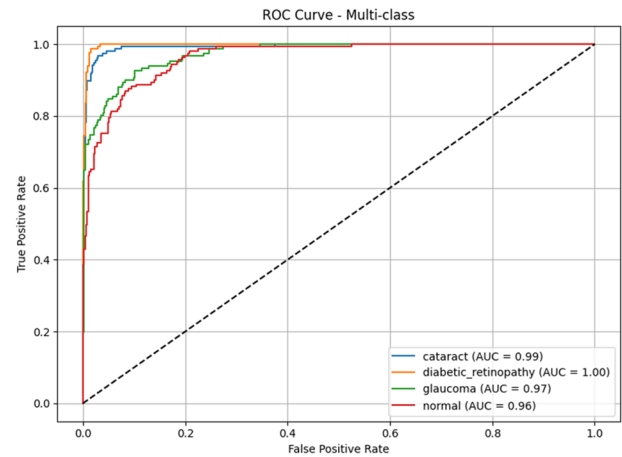
(b). Confusion matrix of DR severity classifier.

In the first stage, the ocular disease classifier demonstrates robust generalization across all four classes, Cataract, Diabetic Retinopathy, Glaucoma, and Normal, with an overall accuracy of 94.15% as presented in Table 4. Notably, the model achieves the highest F1-score for the DR class (0.9790), reflecting excellent precision (0.9702) and recall (0.9879). While performance is slightly lower for Glaucoma (F1-score: 0.9072), precision (0.9429) remains high, indicating some misclassification, primarily in recall (0.8742). The macro-averaged F1-score of 0.9405 and weighted average of 0.9412 confirm consistent performance across balanced class distributions, underscoring the model's efficacy in detecting multiple retinal pathologies. In the second stage, the system refined its diagnostic capability by grading the severity of DR into three clinically relevant categories: Early NPDR, Severe NPDR, and PDR. As shown in Table 5, the model achieved high performance across all classes, with particularly notable results for *Severe NPDR*, where the recall reached 1.000, indicating that all true cases were correctly identified. The overall accuracy of 96.59%, along with high macro and weighted averages, reflects the model's robustness in handling the complexities of DR progression.

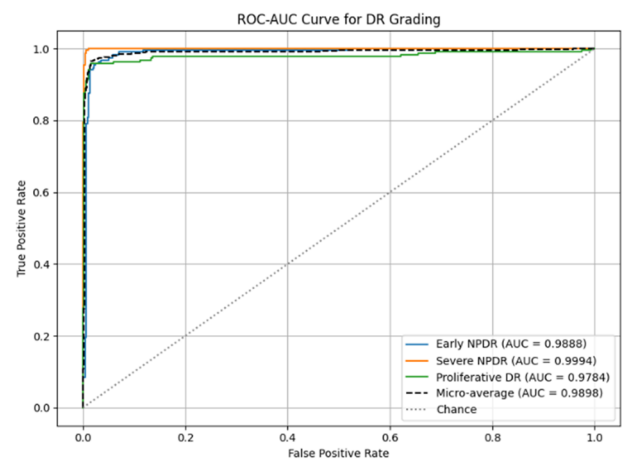
3.3 Confusion matrix

Figure 4(a) illustrates the confusion matrix of the ocular disease classifier. The model demonstrates strong classification capabilities across all four categories. For Cataract, 150 out of 156 samples were correctly classified, with only minor misclassifications into Glaucoma and Normal (3 each), indicating high specificity. The DR class

Fig. 5 (a) ROC curve of ocular disease classifier. **(b)** ROC curve of DR severity classifier



(a). ROC curve of ocular disease classifier.



(b). ROC curve of DR severity classifier.

shows excellent separation, with 160 out of 165 correctly identified and only five misclassified as Normal, affirming the model's sensitivity to DR features.

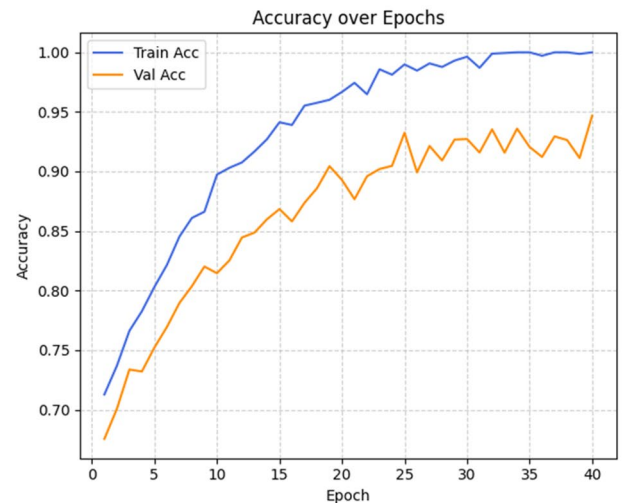
Glaucoma poses the greatest classification challenge, with 119 correct predictions out of 151, while 25 instances were misclassified as Normal and a smaller portion as Cataract and DR. This reflects the class's known clinical overlap and suggests room for improved feature discrimination. The Normal class was accurately identified in 146 out of 161 cases, with most confusion occurring with Glaucoma, which may share subtle optic nerve features. Overall, the confusion matrix confirms the model's robustness but highlights inter-class ambiguity primarily between Glaucoma and Normal, a common challenge in fundus-based diagnostics.

Figure 4(b) presents the confusion matrix for the DR severity classifier, showcasing a high accuracy in grading disease progression. Early NPDR was correctly classified in 211 out of 225 instances, with only a few cases confused with Severe NPDR (6) and PDR (8). Notably, Severe NPDR achieved perfect recall with all 225 cases correctly identified, underscoring the model's precise recognition of mid-stage retinal changes. Proliferative DR (PDR) was accurately identified in 216 of 225 cases, with minor misclassification into the adjacent stages.

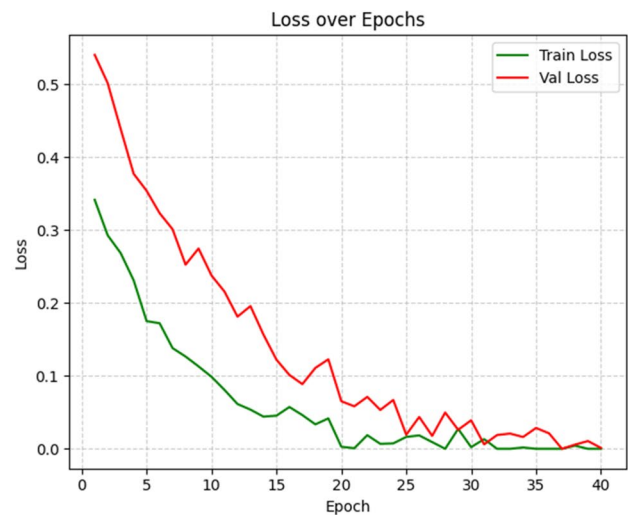
3.4 ROC-AUC curves

To further evaluate the discriminative capacity of the proposed framework, ROC curves were generated for both classification stages, as illustrated in Fig. 5(a) and (b). In Fig. 5(a), the ocular disease classifier demonstrates

Fig. 6 (a) Training and validation accuracy curve of stage 1.
(b) Training and validation loss curve of stage 1



(a). Training and validation accuracy curve of stage 1.



(b). Training and validation loss curve of stage 1.

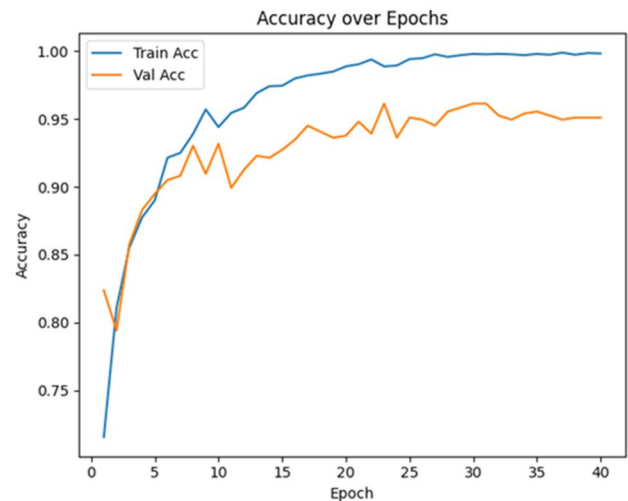
excellent separability among the four classes, Cataract, Diabetic Retinopathy, Glaucoma, and Normal, with Area Under the Curve (AUC) values of 0.99, 1.00, 0.97, and 0.96, respectively. The perfect AUC of 1.00 for DR underscores the model's exceptional capability in identifying diabetic-related pathologies, while the high AUCs across the remaining classes reflect consistent and reliable class discrimination.

In Fig. 5(b), the DR severity grading model maintains similarly high performance, achieving AUC values of 0.9888 for Early NPDR, 0.9994 for Severe NPDR, and 0.9784 for PDR. The near-perfect micro-average AUC of 0.9898 indicates that the model performs robustly across all DR stages, capturing subtle differences in progression severity.

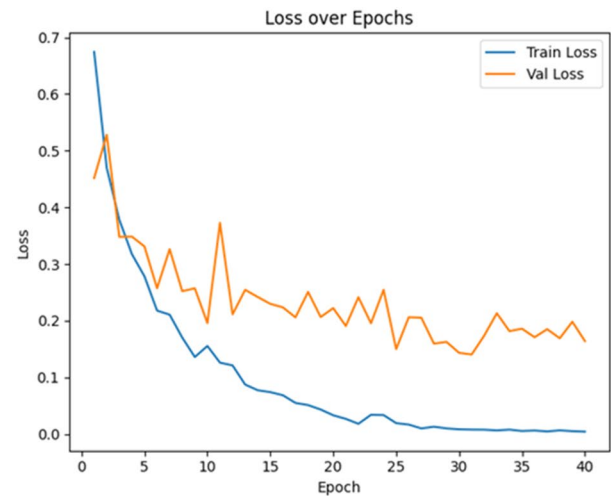
3.5 Training and validation trends

The learning behavior of both stages in the proposed framework was analyzed over 40 training epochs using accuracy and loss curves, as presented in Figs. 6 and 7. In Fig. 6(a–b), the ocular disease classifier also exhibits consistent improvements in both training and validation accuracy, reaching approximately 99% and 94%, respectively. The decreasing loss curves show that both training and validation loss decline steadily.

Fig. 7 (a) Training and validation accuracy curve of stage 2.
(b) Training and validation loss curve of stage 2



(a). Training and validation accuracy curve of stage 2.



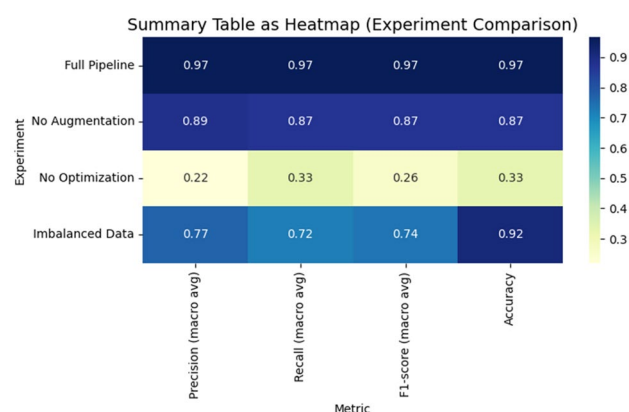
(b). Training and validation loss curve of stage 2.

Table 6 Experimental results of augmentation and optimization effects in ocular disease classification performance

Experiment	Precision	Recall	F1-Score	Accuracy
Full Pipeline	0.9415	0.9404	0.9405	0.9415
No Augmentation	0.9264	0.9238	0.9241	0.9258
No Optimization	0.8434	0.8403	0.8356	0.8452

Table 7 Experimental results of augmentation, optimization, and balancing effects in DR severity grading performance

Experiment	Precision	Recall	F1-Score	Accuracy
Full Pipeline	0.9661	0.9659	0.9658	0.9659
No Augmentation	0.8860	0.8741	0.8736	0.8741
No Optimization	0.2200	0.3333	0.2591	0.3333
Imbalanced Data	0.7705	0.7201	0.7382	0.9158

Fig. 8 Heatmap summary of stage 2 performance effects**Table 8** Performance comparison of models for ocular disease classification (stage 1)

Model	Precision (Macro Avg)	Recall (Macro Avg)	F1-Score (Macro Avg)	Testing Accuracy	Inference Time (s/ image)
ResNet-50	0.8931	0.8839	0.8857	0.8847	0.244750
VGG-16	0.8867	0.8735	0.8753	0.8752	0.314170
DenseNet-121	0.8788	0.8684	0.8708	0.8689	0.255208
ViT	0.9196	0.9198	0.9197	0.9210	0.303462
Swin Transformer	0.9415	0.9404	0.9405	0.9415	0.399532

Table 9 Performance comparison of models for DR severity grading (stage 2)

Model	Precision (Macro Avg)	Recall (Macro Avg)	F1-Score (Macro Avg)	Testing Accuracy	Inference Time (s/ image)
ResNet-50	0.9104	0.9081	0.9081	0.9081	0.108955
VGG-16	0.8882	0.8830	0.8833	0.8830	0.314992
DenseNet-121	0.9188	0.9141	0.9127	0.9141	0.086029
Swin Transformer	0.9450	0.9437	0.9434	0.9437	0.181311
ViT	0.9661	0.9659	0.9658	0.9659	0.302545

Similarly, in Fig. 7(a–b), the DR severity classifier demonstrates stable convergence, with training accuracy steadily increasing and plateauing near 100%, while validation accuracy rises to approximately 95%, indicating strong generalization. The corresponding loss curves show a sharp decline in both training and validation loss, with minimal fluctuations, further confirming the model's robustness and absence of significant overfitting. The alignment between the training and validation curves suggests effective learning dynamics and good model calibration.

4 Discussion

This section provides a detailed interpretation of the experimental results, emphasizing the impact of key components in the proposed pipeline, comparing model performance against various deep learning backbones, and situating the findings within the broader landscape of recent advancements in ocular disease classification and DR grading.

4.1 Ablation studies

To systematically assess the contribution of individual components within the pipeline, ablation studies were conducted at each stage. The results, presented in Tables 6 and 7, highlight the critical roles of data augmentation, model optimization, and class balancing. In the ocular disease classification task (stage 1), removing data augmentation resulted in a modest decline in performance, about 1.5% drop in accuracy and F1-score, indicating that augmentation aids generalization by enhancing input variability. However, the removal of model optimization caused a far more substantial decline in performance, with the F1-score dropping from 0.9405 to 0.8356 and accuracy decreasing from 0.9415 to 0.8452. These results underscore that optimization strategies such as learning rate scheduling, weight decay, and optimizer selection are vital for achieving model convergence and stability.

In stage 2, a similar trend is observed, as summarized in Table 7 and illustrated in Fig. 8. The impact of optimization was even more pronounced. Eliminating optimization resulted in a dramatic decline in model performance, with the F1-score plummeting from 0.9658 to 0.2591 and accuracy dropping to 33.3%. This finding reinforces the necessity of robust optimization strategies, particularly in more complex classification scenarios involving subtle pathological differences.

Additionally, training on the original imbalanced dataset significantly reduced the model's ability to generalize, with precision dropping to 0.7705 and F1-score to 0.7382. These declines indicate that class balancing is essential when dealing with real-world medical datasets that often exhibit skewed distributions.

The absence of augmentation also led to a measurable decline in DR grading performance, further supporting the importance of this component in maintaining generalization. Overall, while each element contributes meaningfully to the pipeline's success, optimization emerges as the most critical, followed by class balancing and augmentation.

4.2 Comparison with backbone deep learning models

Beyond the ablation studies, the effectiveness and generalization capability of the proposed model were evaluated through comparisons with other deep learning backbones. Tables 8 and 9 summarize the results for stage 1 and stage 2, respectively. In stage 1, transformer-based models outperformed traditional CNN architectures as shown in Table 8. The Swin Transformer achieved the highest macro-averaged F1-score of 0.9405 and an overall testing accuracy of 0.9415, slightly surpassing the ViT model. Both transformers significantly outperformed CNN-based models such as ResNet-50, VGG-16, and DenseNet-121, which recorded lower accuracy and F1-scores. These results indicate that transformer architectures, with their self-attention mechanisms and ability to model long-range dependencies, are particularly well-suited for multiclass ocular disease detection, especially when subtle variations in fundus images need to be captured.

In stage 2, where the task was to grade DR severity, the ViT model demonstrated superior performance compared to all other backbones, including the swin transformer. The ViT achieved an accuracy of 96.59% and a macro F1-score of 96.58%, indicating its ability to distinguish between clinically relevant DR stages with high precision and recall. Furthermore, ViT offered competitive inference times, processing images in approximately 0.30 s per image, which suggests its suitability for real-time or near-real-time clinical applications. The

Table 10 Performance comparison of the most recent state-of-the-art methods for DR grading on DDR dataset

Refs.	Year	Methodology	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
[48]	2024	MobileNet	71.00	70.00	71.00	71.00
[40]	2024	ViT + Modified CapsNet	80.36	59.00	55.00	56.00
[49]	2025	CNN	84.15	83.37	84.16	83.43
[50]	2025	CNN	85.77	—	—	—
[51]	2025	CNN + SVM	95.92	95.92	96.02	96.02
Proposed Model	2025	ViT	96.59	96.61	96.59	96.58

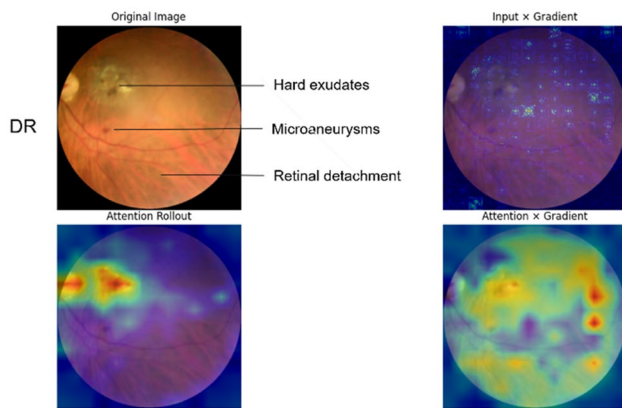


Fig. 9 Comparative visualization of explainability techniques for DR image

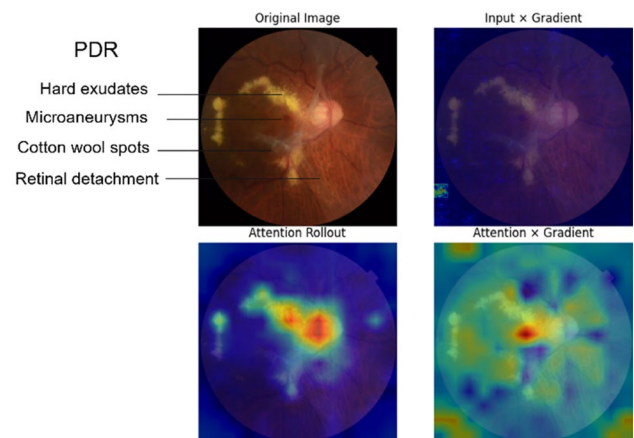


Fig. 10 Comparative visualization of explainability techniques for PDR image

combination of high diagnostic accuracy and computational efficiency makes ViT a promising architecture for deployment in clinical decision support systems.

4.3 Benchmarking against state-of-the-art

To assess the competitiveness of the proposed pipeline on a broader scale, we benchmarked it against several state-of-the-art methods from recent literature on the DDR dataset. As shown in Table 10, our ViT-based method achieved the highest accuracy (96.59%) and F1-score (96.58%), outperforming all listed methods, including different CNN architectures and advanced capsule-based models. These results reinforce the superiority of ViT as a backbone for medical image analysis tasks and confirm the effectiveness of the full pipeline architecture.

5 Model explainability

Explainability plays a critical role in clinical artificial intelligence systems, particularly when diagnostic safety, trust, and transparency are paramount. To ensure the proposed model's decisions are grounded in medically meaningful evidence, interpretability analyses were conducted across both stages of the pipeline: fundus disease classification and diabetic retinopathy severity grading. In each case, visual attribution methods were used to evaluate whether the model's attention aligns with key pathological features.

Figure 9. presents the interpretability results for a case classified as diabetic retinopathy in the first stage. The original fundus image (top-left) displays hallmark pathological signs, including hard exudates, microaneurysms, and indications of retinal detachment. Three explainability techniques were employed to interpret the model's decision. The Input $\times$ Gradient method highlights image regions where small pixel changes would most influence the model's output. While it responds to some abnormal areas, the visualization remains noisy and lacks spatial precision, limiting its diagnostic utility. In contrast, the Attention Rollout map reveals the broader spatial distribution of the model's focus across the image. This method indicates model attention near the fovea and peripheral retina, overlapping with identified lesions, though the output remains relatively coarse. The most precise and clinically relevant results come from the Attention $\times$ Gradient map, which fuses attention weights with gradient sensitivity. This technique produces a sharply localized heatmap that aligns closely with annotated lesions, confirming that the model's decision is based on valid anatomical features.

Figure 10. illustrates a second-stage case diagnosed as proliferative diabetic retinopathy, the most severe and vision-threatening form of the disease. The original image reveals multiple advanced pathological markers, including dense clusters of hard exudates, microaneurysms, cotton wool spots, and signs of retinal detachment—each annotated by a clinical expert. Input $\times$ Gradient again produces sparse and diffuse highlights, offering limited interpretability. The Attention Rollout map provides a more structured focus, capturing attention on areas with cotton wool spots and exudates, along with peripheral zones of retinal detachment. However, the most informative attribution again comes from Attention $\times$ Gradient, which accurately localizes all major lesions with high spatial fidelity. This alignment with clinical findings indicates that the model forms predictions based on appropriate visual cues.

Across both stages, the interpretability results consistently demonstrate that the model's predictions are not only accurate but also grounded in clinically relevant and anatomically consistent regions. Among the evaluated methods, Attention $\times$ Gradient repeatedly provides the most detailed and trustworthy visualizations, making it particularly well-suited for validating model decisions in a medical context. This level of transparency supports clinician trust and facilitates the responsible integration of deep learning models into real-world diagnostic workflows. The model's ability to focus on expert-defined features in both disease classification and severity grading represents a significant step toward explainable and deployable AI in ophthalmology.

6 Conclusion

This paper introduces a novel, explainable AI framework for automated ocular disease classification and DR severity grading, achieving testing accuracies of 94.15% and 96.59%, respectively. The two-stage vision transformer-based model significantly outperforms leading CNN baselines, with macro F1-score improvements of up to 9.3%, highlighting its superior feature extraction and robustness. Ablation studies confirm the critical role of data augmentation and class balancing, with performance declines of up to 6.7% in accuracy and 20.2% in F1-score when these components are removed. The integration of attention-based explainability enhances clinical transparency and trust. Overall, the proposed framework demonstrates high accuracy, robust generalization, and efficient inference, offering a scalable and interpretable solution for early-stage ocular disease screening, particularly valuable in resource-limited healthcare environments. Future work will focus on domain adaptation, lightweight model deployment, and multi-modal data integration to further improve generalization and clinical utility. A key limitation of this study is the lack of external validation on unseen clinical datasets from different geographic or demographic populations, which will be addressed in future studies to ensure broader applicability.

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Data availability The datasets used in this study are publicly available and were downloaded from the Kaggle platform: <https://www.kaggle.com/datasets/gunavenkatdoddi/eye-diseases-classification> and <https://www.kaggle.com/datasets/mariaherrerot/ddrdataset>.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest to report regarding the present study.

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